EE140 Introduction to Communication Systems

Lecture 16

Instructor: Prof. Xiliang Luo

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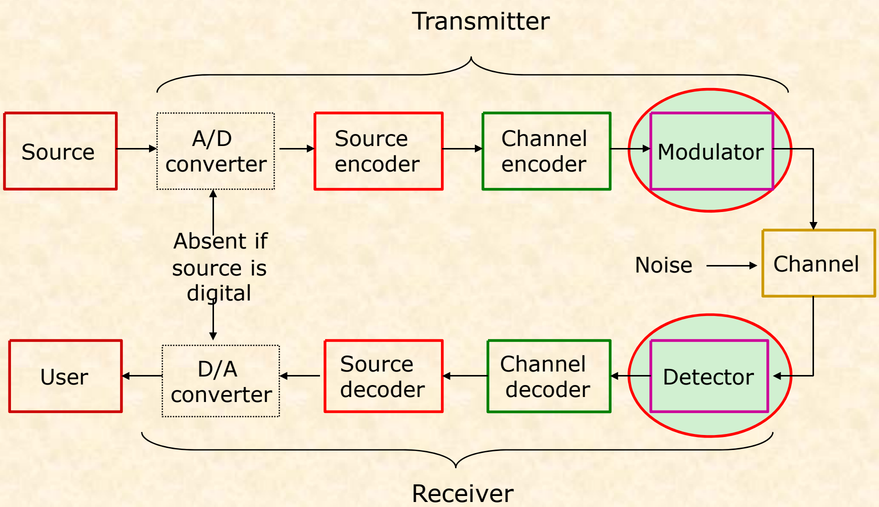
Architecture of a (Digital) Communication System



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Content

- Review: Signal design with zero ISI
- Signal Space

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Signal Design for Bandlimited Channel
Zero ISI

- Nyquist condition for Zero ISI for pulse shape $p(t)$

$$p(nT) = \begin{cases} 1 & n = 0 \quad \textit{Echos made to be zero} \\ 0 & n \neq 0 \quad \textit{at sampling points} \end{cases}$$

$$\text{or } \sum_{k=-\infty}^{\infty} P(f + \tfrac{k}{T}) = T$$

- With the above condition, the receiver output simplifies to

$$v(t_m) = A_m + n_o(t_m)$$

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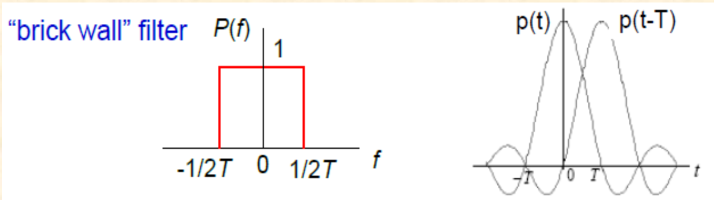
Nyquist Condition: Ideal Case

- Nyquist's first method for eliminating ISI is to use

$$P(f) = \begin{cases} 1 & |f| < \frac{1}{2T} \\ 0 & otherwise \end{cases} \Leftrightarrow p(t) = \frac{\sin(\pi t/T)}{\pi t/T} = sinc\left(\frac{t}{T}\right)$$



- $B_0 = \frac{1}{2T} = \frac{R_S}{2}$ = Nyquist bandwidth
- The minimum transmission bandwidth for zero ISI. A channel with bandwidth B_0 can support a max. transmission rate of $2B_0$ symbols/sec

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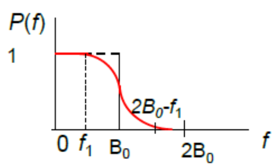
Practical Solution: Raised Cosine Spectrum

- $P(f)$ is made up of 3 parts: passband, stopband, and transition band. The transition band is shaped like a cosine wave.

$$P(f) = \begin{cases} 1 & 0 \le |f| < f_1 \\ \frac{1}{2}\left\{1 + \cos\left[\frac{\pi(|f| - f_1)}{2B_0 - 2f_1}\right]\right\} & f_1 \le |f| < 2B_0 - f_1 \\ 0 & |f| \ge 2B_0 - f_1 \end{cases}$$



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Optimum Transmit/Receive Filter

- Recall that when zero-ISI condition is satisfied by $p(t)$ with raised cosine spectrum $P(f)$, then the sampled output of the receiver filter is $V_m = A_m + N_m$ (assume $p(0)$=1)
- Consider binary PAM transmission: $A_m = \pm d$
- Variance of noise $N_m = \sigma^2 = \int_{-\infty}^{\infty} S_n(f)\,|H_R(f)|^2 df$

with $P(f) = H_T(f)H_C(f)H_R(f)$ $\quad p(t) = h_T(t) * h_C(t) * h_R(t)$

$$\Rightarrow P_e = Q(\frac{d}{\sigma})$$

Error Probability can be minimized through a proper choice of $\mathrm{H_R(f)}$ and $\mathrm{H_T(f)}$ so that d/σ is maximum (assuming $\mathrm{H_C(f)}$ fixed and P(f) given)

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Optimal Solution (cont'd)

- Noise variance at the output of the receive filter is

$$\sigma^2 = \frac{N_0}{2}\int_{-\infty}^{\infty} |H_R(f)|^2 df = \frac{N_0}{2}\int_{-W}^{W} \frac{P(f)}{|H_C(f)|} df$$

$$\Rightarrow P_{e,min} = Q\left[\sqrt{\frac{2\mathrm{E}_{av}}{N_0}\underbrace{\left\{\int_{-W}^{W}\frac{P(f)}{|H_C(f)|}df\right\}^{-1}}_{\text{Performance loss due to channel distortion}}}\right]$$

- Special case: $H_C(f) = 1$ for $|f| \le W$
  - This is the ideal case with "flat" fading
  - No loss, same as the matched filter receiver for AWGN channel

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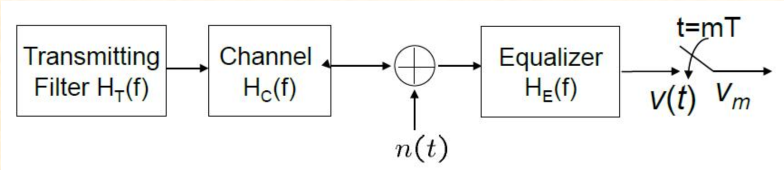
Equalizer Configuration



- Overall frequency response

$$H_o(f) = H_T(f)H_C(f)H_E(f)$$

- Nyquist criterion for zero-ISI

$$\sum_{k=-\infty}^{\infty} H_o\left(f + \frac{k}{T}\right) = constant$$

- Ideal zero-ISI equalizer is an inverse channel filter with

$$H_E(f) \propto \frac{1}{H_T(f)H_C(f)} \qquad |f| \leq 1/2T$$

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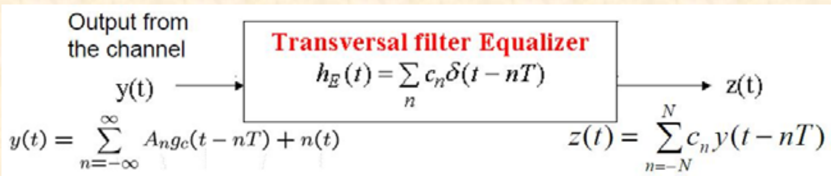
MMSE Criterion



- The output is sampled at $t = mT$:

$$z(mT) = \sum_{n=-N}^{N} c_n y[(m-n)T]$$

- Let $\mathrm{A_m}$ = desired equalizer output

$$MSE \triangleq E\left[(z(mT) - A_m)^2\right] = \text{Minimum}$$

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Content

- Review: Signal design with zero ISI
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Traditional Bandpass Signal Representations

- Baseband signals vs. passband signals
- Passband signals can be represented in three forms
  - Magnitude and Phase representation
  - Quadrature representation
  - Complex Envelop representation

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Magnitude and Phase Representation

- Magnitude and Phase Representation

$$s(t) = a(t)\cos(2\pi f_c t + \vartheta(t))$$

where a(t) is the magnitude of s(t) and θ(t) is the phase of s(t)

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Quadrature or I/Q Representation

- Quadrature or I/Q Representation

$$\begin{aligned} s(t) &= I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t) \\ &= Re\left[(I(t) + jQ(t))e^{2\pi f_c t}\right] \end{aligned}$$

where I(t) and Q(t) are real-valued baseband signals called the in-phase and quadrature components of s(t).

- Signal space is a more convenient way than I/Q representation to study modulation scheme

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Signal Space

- Basic Idea: If a signal can be represented by n-tuple, then it can be treated in much the same way as a n-dim vector.
- Consider a signal x(t) and suppose that

$$x(t) = \sum_{i=1}^{N} x_i \varphi_i(t)$$

- If every signal can be written as above, then $\{\varphi_i(t)\}_{i=1}^{N}$ are the basis functions

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Orthonormal Basis

- Signal set $\{\varphi_i(t)\}_{i=1}^{N}$ is an orthonormal set if

$$\langle \varphi_i(t), \varphi_j(t) \rangle = \int_{-\infty}^{\infty} \varphi_i(t)\varphi_j(t)dt = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

  - Orthogonal, if not normalized to 1.
- Vector representation of x(t)

$$x(t) = \sum_{i=1}^{N} x_i \varphi_i(t)$$

$$\Downarrow$$

$$(x_1, x_2, \ldots, x_N)$$

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Basis Functions for a Signal Set

- Consider a set of M signals (M-ary symbol) $\{s_i(t)\}_{i=1}^{M}$ with finite energy. That is

$$\int_{-\infty}^{\infty} s_i^2(t)dt < \infty$$

- Then, we can express each of these waveforms as weighted linear combination of orthonormal signals

$$s_i(t) = \sum_{j=1}^{N} s_{ij}\varphi_j(t)$$

where N ≤ M is the dimension of the signal space, and $\{\varphi_j(t)\}_{j=1}^{N}$ are called the orthonormal basis functions

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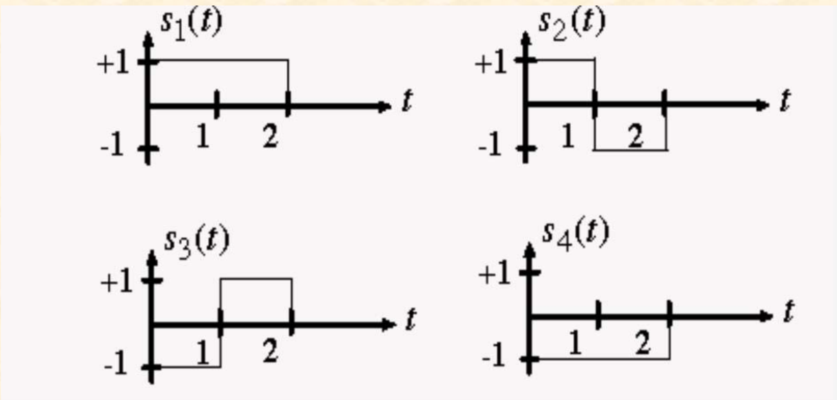
Example 1

- Consider the following signal set:



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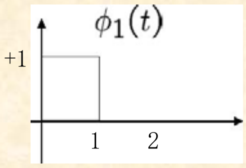
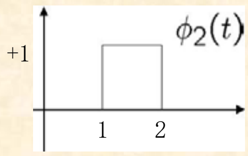
Example 1 (cont'd)

- By inspection, the signals can be expressed in terms of the following two basis functions:





- We have

$$s_1(t) = 1 \cdot \varphi_1(t) + 1 \cdot \varphi_2(t) \qquad s_2(t) = 1 \cdot \varphi_1(t) - 1 \cdot \varphi_2(t)$$

$$s_3(t) = -1 \cdot \varphi_1(t) + 1 \cdot \varphi_2(t) \qquad s_4(t) = -1 \cdot \varphi_1(t) - 1 \cdot \varphi_2(t)$$

- Question: $\{\varphi_j(t)\}_{j=1}^{N}$ orthonormal basis functions?

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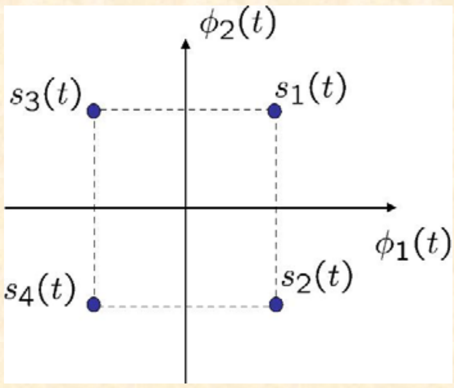
Example 1 (cont'd)

- Constellation diagram
  - A representation of a digital modulation scheme in the signal space
  - Axes are labeled with the basis functions $\varphi_1(t)$ and $\varphi_2(t)$
  - Possible signals are plotted as points, called constellation points



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Example 2

- Suppose the signal set can be represented in the following form

$$s(t) = \pm cos(2\pi f_c t) \pm sin(2\pi f_c t)$$

with $t \in [0, T)$, and $f_c T \gg 1$

- Basis functions

$$\varphi_1(t) = \sqrt{\frac{2}{T}} cos(2\pi f_c t) \qquad \varphi_2(t) = \sqrt{\frac{2}{T}} sin(2\pi f_c t)$$

with $t \in [0, T)$

- Orthonomality?

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Example 2 (cont'd)

- Orthonormality

$$\int_0^T \varphi_1(t)\varphi_2(t)dt = \int_0^T \sqrt{\frac{2}{T}} cos(2\pi f_c t)\sqrt{\frac{2}{T}} sin(2\pi f_c t)dt$$

$$= \frac{2}{T}\int_0^T \frac{1}{2}[sin(4\pi f_c t) + sin(0)]dt$$

$$= -\frac{1}{4\pi f_c T}[cos(4\pi f_c t)]_0^T \approx 0 \qquad f_c T >> 1$$

$$\int_0^T \varphi_1^2(t)dt = \int_0^T \varphi_2^2(t)dt = \frac{2}{T}\int_0^T cos^2(2\pi f_c t)dt$$

$$= \frac{2}{T}\int_0^T \frac{1}{2}[cos(4\pi f_c t) + 1]dt$$

$$= 1 + \frac{1}{4\pi f_c T}[sin(4\pi f_c t)]_0^T \approx 1 \qquad f_c T >> 1$$

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Notes on Signal Space

- Two entirely different signal sets can have the same geometric representation.
- The underlying geometry will determine the performance and the receiver structure

- Next question: is there any method to find a complete orthonormal basis for an arbitrary signal set?
  - Gram-Schmidt Orthogonalization (GSO) procedure

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Gram Schmidt Orthogonalization (GSO) Procedure

- Suppose a signal set is given

$$\{s_i(t)\}_{i=1}^{M}$$

- Find the orthogonal basis functions for this signal set

$$\{\varphi_j(t)\}_{j=1}^{N}$$

where $N \leq M$

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Step 1: Construct the First Basis Function

- Compute the energy in signal 1:

$$E_1 = \int_{-\infty}^{\infty} s_1^2(t)dt$$

- The first basis function is just a normalized version of $s_1(t)$

$$\varphi_1(t) = \frac{1}{\sqrt{E_1}} s_1(t)$$

- Projection of signal to signal space

$$s_1(t) = s_{11}\varphi_1(t) \Rightarrow s_{11} = \sqrt{E_1}$$

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Step 2: Construct the Second Basis Function

- Compute correlation between signal 2 and basis function 1

$$s_{21} = \int_{-\infty}^{\infty} s_2(t)\varphi_1(t)dt$$

- Subtract off the correlation portion

$$g_2(t) = s_2(t) - s_{21}\varphi_1(t)$$

$g_2(t)$ is orthogonal to $\varphi_1(t)$

- Compute the energy in the remaining portion

$$E_{g_2} = \int_{-\infty}^{\infty} g_2^2(t)dt$$

- Normalize the remaining portion $\varphi_2(t) = \frac{1}{\sqrt{E_{g_2}}} g_2(t)$
- Projection

$$s_{21} = \int_{-\infty}^{\infty} s_2(t)\varphi_1(t)dt$$

$$s_{22} = \int_{-\infty}^{\infty} s_2(t)\varphi_2(t)dt = \sqrt{E_{g_2}}$$

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Step 3: Construct Successive Basis Functions

- Iterative process: For signal $s_k(t)$, compute

$$s_{ki} = \int_{-\infty}^{\infty} s_k(t)\varphi_i(t)dt, \text{for } i < k$$

- Define $g_k(t)$

$$g_k(t) = s_k(t) - \sum_{i=1}^{k-1} s_{ki}\varphi_i(t)$$

- Energy

$$E_{g_k} = \int_{-\infty}^{\infty} g_k^2(t)dt$$

- New basis function $\varphi_k(t) = \frac{1}{\sqrt{E_{g_k}}} g_k(t)$

- Projection $s_{kk} = \int_{-\infty}^{\infty} s_k(t)\varphi_k(t)dt = \sqrt{E_{g_k}}$

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Summary of GSO Procedure

- 1st basis function is normalized version of the first signal
- Successive basis functions are found by removing portions of signals that are correlated to previous basis functions and normalizing the result
- This procedure is repeated until all basis functions are found
- If $g_{k(t)} = 0$, no new basis functions is added
- The order in which signals are considered can be arbitrary

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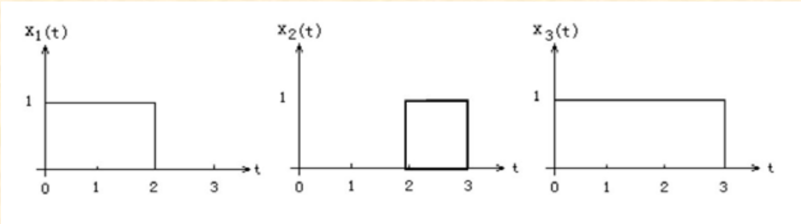
Example: GSO

1. Use the Gram-Schmidt procedure to find a set orthonormal basis functions corresponding to the signals show below



2. Express x_1, x_2, and x_3 in terms of the orthonormal basis functions found in part 1)
3. Draw the constellation diagram for this signal set

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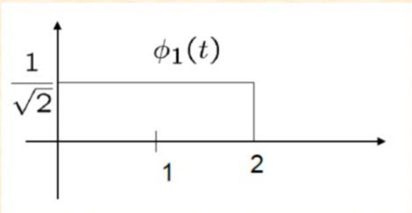
Solution: 1)

- Basis function 1:

$$E_1 = \int_{-\infty}^{\infty} x_1^2(t)dt = 2$$

$$\varphi_1(t) = \frac{1}{\sqrt{2}} x_1(t)$$

$$x_{11} = \sqrt{2}$$



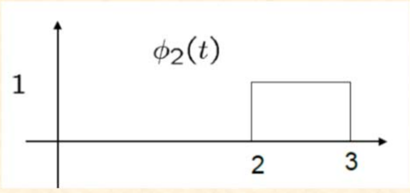
- Basis function 2:

$$x_{21} = \int_{-\infty}^{\infty} x_2(t)\varphi_1(t)dt = 0$$

$$g_2(t) = x_2(t) \quad E_{g_2} = 1$$

$$\varphi_2(t) = x_2(t)$$

$$x_{22} = 1$$



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Solution: 1) (cont'd)

- Basis function 3:

$$x_{31} = \int_{-\infty}^{\infty} x_3(t)\varphi_1(t)dt = \sqrt{2}$$

$$x_{32} = \int_{-\infty}^{\infty} x_3(t)\varphi_2(t)dt = 1$$

$$g_3(t) = x_3(t) - x_{31}\varphi_1(t) - x_{32}\varphi_2(t) = 0$$

No more new basis functions!

- Orthonormal function set

$$\varphi_1(t) = \frac{1}{\sqrt{2}}x_1(t) \qquad \varphi_2(t) = x_2(t)$$

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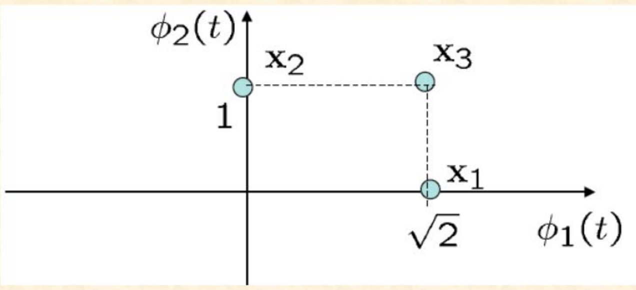
Solution: 2) and 3)

- Express x_1, x_2, x_3 in basis functions

$$x_1(t) = \sqrt{2}\varphi_1(t) \qquad x_2(t) = \varphi_2(t) \qquad x_3(t) = \sqrt{2}\varphi_1(t) + \varphi_2(t)$$

- Constellation diagram



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Exercise

- Given a set of signals (8PSK modulation)

$$s_i(t) = A\cos\left(2\pi f_c t + \frac{\pi}{4} i\right), i = 0,1,\ldots,7, \text{and } t \in [0,T)$$

  - Find the orthonormal basis functions using Gram Schmidt procedure
  - What is the dimension of the resulting signal space?
  - Draw the constellation diagram of this signal set

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Notes on GSO Procedure

- A signal set may have many different sets of basis functions
- A change of basis functions is essentially a rotation of the signal points around the origin.
- The order in which signals are used in the GSO procedure affects the resulting basis functions
- The choice of basis functions does not affect the performance of the modulation scheme

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Thanks for your kind attention!

Questions?

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